Topic: Polynomial Representation and Addition Self-Learning

Q. Explain how a Singly Linked List can be used to represent a polynomial. Describe the algorithm to add two polynomials represented using linked lists with a suitable example.

> **Definition to Remember**

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Polynomial Representation & Addition (Self-Learning)

Each node in the linked list represents one term and is divided into three parts:

1. **Co

Structure in C:

```c

**Example Diagram for 5x^3 + 4x^1 + 2x^0:**

```text

### 2. Polynomial Addition Algorithm

To add `Poly1` and `Poly2`, traverse both lists simultaneously using pointers `p1` and `p2`. Compare their exponents:

- **Case 1: `p1->exponent > p2->exponent`**

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*After the loop, append any remaining terms from either `p1` or `p2` to the Result list.*

### 3. Example Trace

- **Poly1:** 3x^2 + 5x + 2

- **P

## 4. Advantages

- **Memory Efficiency:** Only non-zero terms are stored. (An array for x^{100} + 1 requires 101 slots; a linked list requires only 2 nodes).

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> **Must-Write Points (for 10 marks)**

> 1. Ar

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> **Quick Recall**

> `Poly

(add c